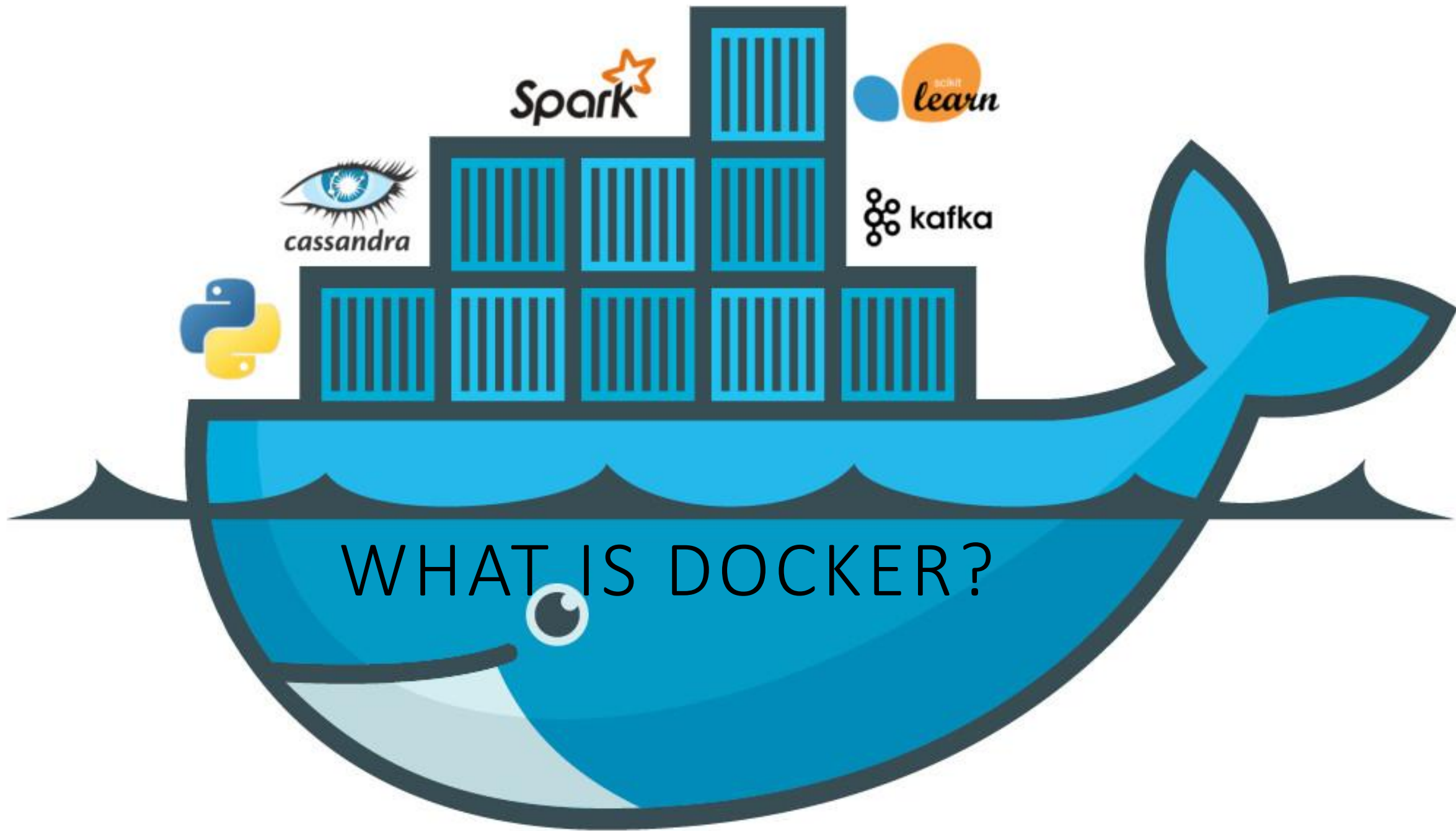






AGENDA

INTRODUCTION TO DOCKER

UNDERSTANDING DOCKER

USE CASES FOR DOCKER

LET'S DO IT TOGETHER NOW.

Q&A

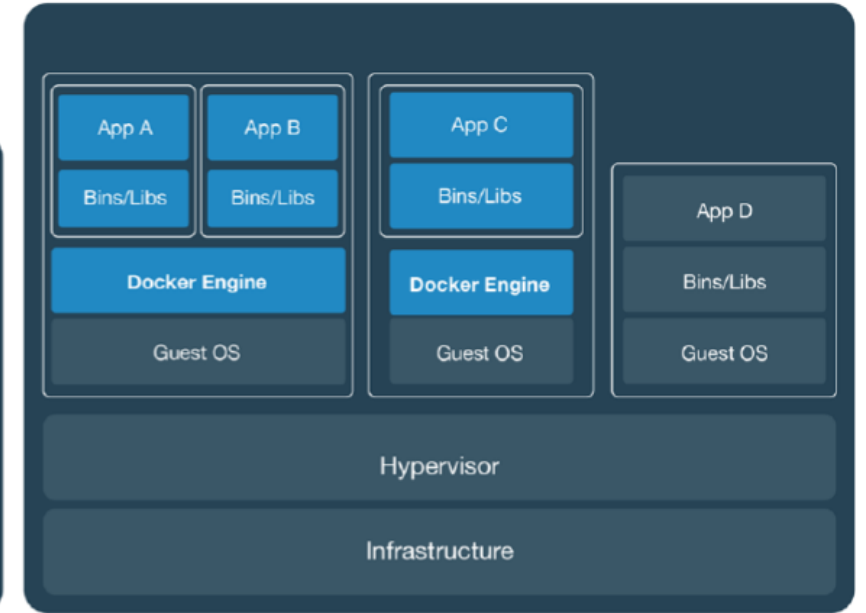
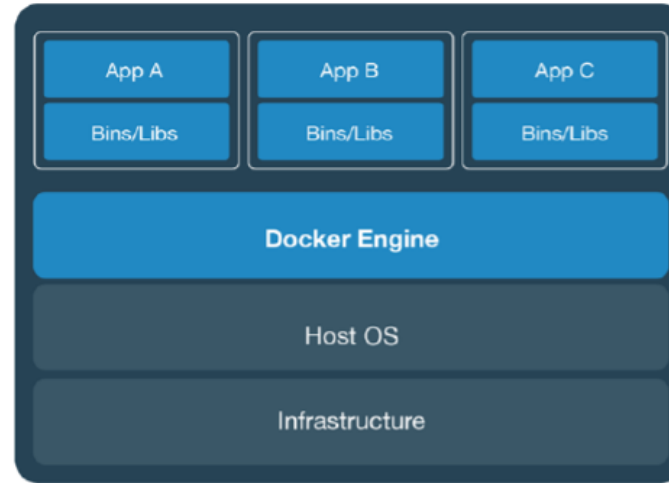


THE POWER OF DOCKER

WHAT IS DOCKER?

Containers Vs VMs

Docker is the ability to run multiple applications on the same hardware in a virtualized environment without over saturating your hardware resources



Docker containers are ethereal.

Docker containers scale to your usage.

Docker allows you to run applications that are OS specific in non OS specific environments.

WHY DO I CARE ABOUT
DOCKER?

WHAT'S IN IT FOR ME?

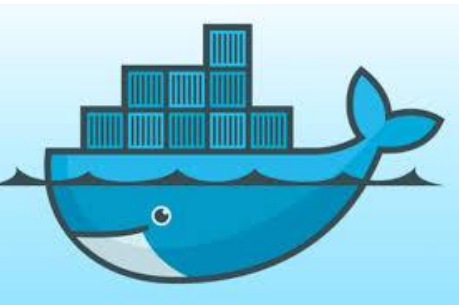




DOCKER USES CASES FOR MAKERS

- All this hardware, and I have to install an OS?
- IoT and Home Automation projects.
- 3D printer farm management.
- Shared resourcing for Astrophotography with PixInsight or Siril
- Telescope control automation without exposing hardware.
- Astronomy simulations.
- Collaborative Data collection, processing, and monitoring.





DOCKER CONTAINER USES CONTINUED

- Web Hosting Solutions – Wiki for exploratory note taking or project note taking
- Network management solutions
- SMB Controllers (more projects, less hardware)
- Hardware vs Software – Remotely manage your hardware without having it directly connected.
- Public Accessibility with CloudFlare Zero Trust env
- LLM for local AI
- Data management and Graphing
- OS trials – Wanna try a new Linux OS?
- Project doesn't support your OS?
- Your project uses software that IS an OS?
- Collaborate on projects together and host a single source of truth for updates from each other.
- Dispose of projects you've abandoned and free up space.
- Flexibility to try more things with lower risk
- Faster deployments and uses

DOCKER IMAGES. WHAT ARE DOCKER IMAGES?

Docker images are precompiled operating system images that have the applications installed at the OS level as a service or daemon.

It's not magic.
It's not a black box.

LET'S DEPLOY SOMETHING WITH DOCKER!

- Deploying Ollama (LLM AI) with Docker!
- Prereqs:
Have docker desktop installed.
- Navigate to the official Ollama site for instructions.

<https://ollama.com/blog/ollama-is-now-available-as-an-official-docker-image>

- Follow instructions to pass GPU to Ollama docker image. (Where applicable.)

Seek feedback

Reflect on performance

Explore new techniques

Set personal goals

Iterate and adapt



THANK YOU

Q/A Time.